



Mathematical Statistics

Homework V



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Casella and Berger 11.32

Observations $(x_i, Y_i), i = 1, 2, \dots, n$, are made according to the model

$$Y_i = \alpha + \beta x_i + \epsilon_i,$$

where $x_1, x_2, \dots, x_n$ are fixed constants and $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ are iid $n(0, \sigma^2)$. The model is then reparameterized as

$$Y_i = \alpha' + \beta'(x_i - \bar{x}) + \epsilon_i.$$

Let $\hat{\alpha}$ and $\hat{\beta}$ denote the MLEs of α and β , respectively, and $\hat{\alpha}'$ and $\hat{\beta}'$ denote the MLEs of α' and β' , respectively.

- (a) Show that $\hat{\beta}' = \hat{\beta}$.
- (b) Show that $\hat{\alpha}' \neq \hat{\alpha}$. In fact, show that $\hat{\alpha}' = \bar{Y}$. Find the distribution of $\hat{\alpha}'$.
- (c) Show that $\hat{\alpha}'$ and $\hat{\beta}'$ are uncorrelated and, hence, independent under normality.

Solution

(a) Under the first parameterization of the model, $Y_i \sim n(\alpha + \beta x_i, \sigma^2)$. Thus, the log-likelihood function for the observations is

$$\mathcal{L}_1 = n \log \left(\frac{1}{\sqrt{2\pi}\sigma} \right) - \frac{1}{2\sigma^2} \sum (y_i - \alpha - \beta x_i)^2. \quad (1)$$

The solution of the equation

$$\frac{\partial \mathcal{L}_1}{\partial \beta} = \frac{1}{\sigma^2} \sum (y_i - \alpha - \beta x_i) x_i \quad (2)$$

$$= 0 \quad (3)$$

is

$$\hat{\beta} = \frac{\sum (y_i - \hat{\alpha}) x_i}{\sum x_i^2}. \quad (4)$$

We can solve for $\hat{\alpha}$ (the MLE of α) in order to write $\hat{\beta}$ (the MLE of β) in terms of only the x_i and y_i . The solution to

$$\frac{\partial \mathcal{L}_1}{\partial \alpha} = \frac{1}{\sigma^2} \sum (y_i - \alpha - \beta x_i) \quad (5)$$

$$= 0 \quad (6)$$

is $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$. So we can write

$$\hat{\beta} = \frac{\sum (y_i - \bar{y} + \hat{\beta}\bar{x})x_i}{\sum (x_i - \bar{x})x_i} \quad (7)$$

Solving the above equation for $\hat{\beta}$ yields

$$\hat{\beta} = \frac{\sum (y_i - \bar{y})x_i}{\sum (x_i - \bar{x})x_i} \quad (8)$$

$$= \frac{\sum (y_i - \bar{y})(x_i - \bar{x} + \bar{x})}{\sum (x_i - \bar{x})(x_i - \bar{x} + \bar{x})} \quad (9)$$

$$= \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) + \bar{x} \sum (y_i - \bar{y})}{\sum (x_i - \bar{x})^2 + \bar{x} \sum (x_i - \bar{x})} \quad (10)$$

$$(11)$$

Since the sum of deviations from the mean is 0, we have

$$\hat{\beta} = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \quad (12)$$

$$= \frac{S_{xy}}{S_{xx}}. \quad (13)$$

Now, under the second parameterization of the model, $Y_i \sim n(\alpha' + \beta'(x_i - \bar{x}), \sigma^2)$, so the log-likelihood function in this case is

$$\mathcal{L}_2 = n \log \left(\frac{1}{\sqrt{2\pi}\sigma} \right) - \frac{1}{2\sigma^2} \sum [y_i - \alpha' - \beta'(x_i - \bar{x})]^2. \quad (14)$$

The solution of the equation

$$\frac{\partial \mathcal{L}_2}{\partial \beta'} = \frac{1}{\sigma^2} \sum [y_i - \alpha' - \beta'(x_i - \bar{x})](x_i - \bar{x}) \quad (15)$$

$$= \frac{1}{\sigma^2} \left[\sum y_i(x_i - \bar{x}) - \alpha' \sum (x_i - \bar{x}) - \beta' \sum (x_i - \bar{x})^2 \right] \quad (16)$$

$$= \frac{1}{\sigma^2} \left[\sum y_i(x_i - \bar{x}) - \beta' \sum (x_i - \bar{x})^2 \right] \quad (17)$$

$$= 0 \quad (18)$$

is

$$\hat{\beta}' = \frac{\sum y_i(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \quad (19)$$

$$= \frac{\sum (y_i - \bar{y} + \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \quad (20)$$

$$= \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) + \bar{y} \sum (x_i - \bar{x})}{\sum (x_i - \bar{x})^2}. \quad (21)$$

Since the sum of deviations from the mean is 0, we have

$$\hat{\beta}' = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \quad (22)$$

$$= \frac{S_{xy}}{S_{xx}}. \quad (23)$$

Therefore, $\hat{\beta} = \hat{\beta}'$.

(b) In part **(a)**, we found that $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$. What remains is to show that $\hat{\alpha}' = \bar{y}$. Using $\mathcal{L}_2$ as defined in part **(a)**, we can find $\hat{\alpha}'$ by solving

$$\frac{\partial \mathcal{L}_2}{\partial \alpha'} = \frac{1}{\sigma^2} \sum [y_i - \alpha' - \beta'(x_i - \bar{x})] \quad (24)$$

$$= \frac{1}{\sigma^2} \sum (y_i - \alpha') - \frac{\beta'}{\sigma^2} \sum (x_i - \bar{x}) \quad (25)$$

$$= \frac{1}{\sigma^2} \sum y_i - \frac{n\alpha'}{\sigma^2} \quad (26)$$

$$= 0 \quad (27)$$

for α' . The solution of this equation is clearly

$$\hat{\alpha}' = \frac{1}{n} \sum y_i \quad (28)$$

$$= \bar{y}. \quad (29)$$

Thus, $\hat{\alpha} \neq \hat{\alpha}'$.

Now, since the $Y_i \sim n(\alpha' + \beta'(x_i - \bar{x}), \sigma^2)$, we have

$$\bar{Y} \sim n\left(\frac{1}{n} \sum \alpha' + \frac{\beta'}{n} \sum (x_i - \bar{x}), \frac{\sigma^2}{n}\right). \quad (30)$$

But $\frac{1}{n} \sum \alpha' = \alpha'$ and $\frac{\beta'}{n} \sum (x_i - \bar{x}) = 0$. Thus,

$$\bar{Y} \sim n\left(\alpha', \frac{\sigma^2}{n}\right). \quad (31)$$

(c) By Theorem 11.3.3 (Casella and Berger § 11.3), the covariance of $\hat{\alpha}'$ and $\hat{\beta}'$ is

$$\text{Cov}(\hat{\alpha}', \hat{\beta}') = \left(\frac{-\sigma^2}{S_{xx}}\right) \frac{1}{n} \sum (x_i - \bar{x}) \quad (32)$$

$$= 0. \quad (33)$$

Since the covariance of $\hat{\alpha}'$ and $\hat{\beta}'$ is 0, their correlation is also 0.

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Casella and Berger 11.33

Observations (X_i, Y_i) , $i = 1, 2, \dots, n$, are made from a bivariate normal population with parameters $(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$, and the model $Y_i = \alpha + \beta x_i + \epsilon_i$ is going to be fit.

(a) Argue that the hypothesis $H_0 : \beta = 0$ is true if and only if the hypothesis $H_0 : \rho = 0$ is true.

(b) Show algebraically that

$$\frac{\hat{\beta}}{S/\sqrt{S_{xx}}} = \sqrt{n-2} \frac{r}{\sqrt{1-r^2}},$$

where r is the sample correlation coefficient, the MLE of ρ .

(c) Show how to test $H_0 : \rho = 0$, given only r^2 and n , using Student's t with $n - 2$ degrees of freedom

Solution

(a) Under the bivariate normal model, $\beta = \rho \frac{\sigma_Y}{\sigma_X}$ (Casella and Berger § 11.3). Assuming $\sigma_Y \neq 0$, it is clear that if $\beta = 0$, so does ρ , and vice-versa.

(b) We begin with the facts that $\hat{\beta} = \frac{S_{xy}}{S_{xx}}$ and $r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}$. We can write

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} \tag{34}$$

$$= \frac{S_{xy} \sqrt{S_{yy}}}{\sqrt{S_{xx}} \sqrt{S_{xx}} \sqrt{S_{yy}}} \tag{35}$$

$$= r \sqrt{\frac{S_{yy}}{S_{xx}}}. \tag{36}$$

Therefore,

$$\frac{\hat{\beta}}{1/\sqrt{S_{xx}}} = \hat{\beta} \sqrt{S_{xx}} \tag{37}$$

$$= r \sqrt{S_{yy}}. \tag{38}$$

Note that we can write

$$S = \frac{\sqrt{S_{yy} - \hat{\beta} S_{xy}}}{\sqrt{n-2}}.$$

Thus,

$$\frac{\hat{\beta}}{S/\sqrt{S_{xx}}} = r \frac{\sqrt{S_{yy}}}{S} \quad (39)$$

$$= r \frac{\sqrt{S_{yy}}}{\sqrt{S_{yy} - \hat{\beta} S_{xy}}} \sqrt{n-2} \quad (40)$$

$$= \sqrt{n-2} \frac{r}{\sqrt{1 - \hat{\beta} \frac{S_{xy}}{S_{yy}}}} \quad (41)$$

$$= \sqrt{n-2} \frac{r}{\sqrt{1 - \frac{S_{xy}^2}{S_{xx} S_{yy}}}} \quad (42)$$

$$= \sqrt{n-2} \frac{r}{\sqrt{1-r^2}}. \quad (43)$$

(c) From part (a), we know that testing $H_0 : \rho = 0$ is equivalent to testing $H'_0 : \beta = 0$. From Casella and Berger § 11.3, we know that we reject $H'_0 : \beta = 0$ at significant level α if

$$\left| \frac{\hat{\beta} - 0}{S/\sqrt{S_{xx}}} \right| > t_{n-2, \alpha/2}.$$

But from part (b), we know that $\hat{\beta} = \sqrt{n-2} \frac{r}{\sqrt{1-r^2}} \frac{S}{\sqrt{S_{xx}}}$. Thus, we can reject H'_0 whenever

$$\left| \frac{\hat{\beta} - 0}{S/\sqrt{S_{xx}}} \right| = \left| \frac{\sqrt{n-2} \frac{r}{\sqrt{1-r^2}} \frac{S}{\sqrt{S_{xx}}}}{S/\sqrt{S_{xx}}} \right| \quad (44)$$

$$= \left| \sqrt{n-2} \frac{r}{\sqrt{1-r^2}} \right| \quad (45)$$

$$> t_{n-2, \alpha/2}. \quad (46)$$

But rejecting H'_0 is equivalent to rejecting H_0 . Therefore, we reject $H_0 : \rho = 0$ whenever

$$\left| \sqrt{n-2} \frac{r}{\sqrt{1-r^2}} \right| > t_{n-2, \alpha/2}.$$

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Observations $Y_1, Y_2, \dots, Y_n$ are described by the relationship $Y_i = \theta x_i^2 + \epsilon_i$, where $x_1, x_2, \dots, x_n$ are fixed constants and $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ are iid $n(0, \sigma^2)$.

- (a) Find the least squares estimator of θ .
- (b) Find the MLE of θ .
- (c) Find the best unbiased estimator of θ .

Solution

- (a) To find the least squares estimator of θ , we need to minimize the residual sum of squares

$$\text{RSS} = \sum (y_i - \theta x_i^2)^2. \quad (47)$$

We do this by solving the equation

$$\frac{d}{d\theta} (\text{RSS}) = 2 \sum (y_i - \theta x_i^2) x_i^2 \quad (48)$$

$$= 2 \sum y_i x_i^2 - 2\theta \sum x_i^4 \quad (49)$$

$$= 0 \quad (50)$$

for θ . It is easy to see that the solution is

$$\tilde{\theta} = \frac{\sum y_i x_i^2}{\sum x_i^4},$$

which is the least squares estimator of θ .

- (b) The distribution of each Y_i is $n(\theta x_i^2, \sigma^2)$, so the likelihood function for the observations is

$$L(\theta|\mathbf{y}) = \prod \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{1}{2\sigma^2} (y_i - \theta x_i^2)^2 \right\} \quad (51)$$

$$= \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum (y_i - \theta x_i^2)^2 \right\}. \quad (52)$$

The log-likelihood function is

$$\mathcal{L}(\theta|\mathbf{y}) = n \log \left(\frac{1}{\sqrt{2\pi}\sigma} \right) - \frac{1}{2\sigma^2} \sum (y_i - \theta x_i^2)^2. \quad (53)$$

The solution of

$$\frac{d}{d\theta} \mathcal{L}(\theta|\mathbf{y}) = -\frac{1}{\sigma^2} \sum (y_i - \theta x_i^2) x_i^2 \quad (54)$$

$$= 0 \quad (55)$$

is

$$\hat{\theta} = \frac{\sum y_i x_i^2}{\sum x_i^4},$$

which is the MLE of θ .

(c) The variance of $\hat{\theta}$ is

$$\text{Var}(\hat{\theta}) = \frac{\sigma^2}{[\sum x_i^4]^2} \sum x_i^4 \quad (56)$$

$$= \frac{\sigma^2}{\sum x_i^4}. \quad (57)$$

Let $\tau(\theta) = \theta$. The Crámer-Rao Lower Bound for variances of estimators of θ is

$$\text{CRLB} = \frac{\left(\frac{\partial}{\partial \theta} \tau(\theta)\right)^2}{-E\left[\frac{\partial^2}{\partial \theta^2} \mathcal{L}(\theta|\mathbf{y})\right]} \quad (58)$$

$$= \frac{1}{-E\left[\frac{\partial^2}{\partial \theta^2} \mathcal{L}(\theta|\mathbf{y})\right]}, \quad (59)$$

where

$$\mathcal{L}(\theta|\mathbf{y}) = n \log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{1}{2\sigma^2} \sum (y_i - \theta x_i^2). \quad (60)$$

We have

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{1}{\sigma^2} \sum (y_i - \theta x_i^2) x_i^2 \quad (61)$$

$$= \frac{1}{\sigma^2} \sum y_i x_i^2 - \frac{\theta}{\sigma^2} \sum x_i^4. \quad (62)$$

Therefore,

$$\frac{\partial^2 \mathcal{L}}{\partial \theta^2} = -\frac{1}{\sigma^2} \sum x_i^4. \quad (63)$$

So $-E\left[\frac{\partial^2 \mathcal{L}}{\partial \theta^2}\right] = \frac{1}{\sigma^2} \sum x_i^4$, and

$$\text{CRLB} = \frac{1}{\frac{1}{\sigma^2} \sum x_i^4} \quad (64)$$

$$= \frac{\sigma^2}{\sum x_i^4} \quad (65)$$

$$= \text{Var}(\hat{\theta}). \quad (66)$$

Thus, $\hat{\theta} = \tilde{\theta}$ is the best unbiased estimator of θ .

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Observations $Y_1, Y_2, \dots, Y_n$ are made according to the model $Y_i = \alpha + \beta x_i + \epsilon_i$, where $x_1, x_2, \dots, x_n$ are fixed constants and $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ are iid $n(0, \sigma^2)$. Let $\hat{\alpha}$ and $\hat{\beta}$ denote MLEs of α and β .

- (a) Assume that $x_1, x_2, \dots, x_n$ are observed values of iid random variables $X_1, X_2, \dots, X_n$ with distribution $n(\mu_X, \sigma_X^2)$. Prove that when we take expectations over the joint distribution of X and Y , we still get $E[\hat{\alpha}] = \alpha$ and $E[\hat{\beta}] = \beta$.
- (b) The phenomenon of part (a) does not carry over to the covariance. Calculate the unconditional covariance of $\hat{\alpha}$ and $\hat{\beta}$ (using the joint distribution of X and Y).

Solution

- (a) By the Law of Iterated Expectations, we can write

$$E[\hat{\beta}] = E[E[\hat{\beta}|\mathbf{X}]]. \quad (67)$$

In the case where we are given $\bar{X}$, we know that $\hat{\beta}$ is an unbiased estimator of β . Therefore, $E[\hat{\beta}|\bar{X}] = \beta$. Thus,

$$E[\hat{\beta}] = E[\beta] \quad (68)$$

$$= \beta. \quad (69)$$

Using the Law of Iterated Expectations and the fact that $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$, we have

$$E[\hat{\alpha}] = E[\bar{Y} - \hat{\beta}\bar{X}] \quad (70)$$

$$= E[E[\bar{Y} - \hat{\beta}\bar{X}|\mathbf{X}]] \quad (71)$$

$$= E[E[\bar{Y}|\mathbf{X}] - E[\hat{\beta}\bar{X}|\mathbf{X}]] \quad (72)$$

$$= E[E[\bar{Y}|\mathbf{X}] - \beta\bar{X}] \quad (73)$$

The conditional distribution of $\bar{Y}$ given $\bar{X}$ is $n(\alpha + \beta\bar{x}, \sigma^2)$. Thus, $E[\bar{Y}|\bar{X}] = \alpha + \beta\bar{x}$. So

$$E[\hat{\alpha}] = E[E[\bar{Y}|\mathbf{X}] - \beta\bar{X}] \quad (74)$$

$$= E[\alpha + \beta\bar{X} - \beta\bar{X}] \quad (75)$$

$$= E[\alpha] \quad (76)$$

$$= \alpha. \quad (77)$$

- (b) Using the Law of Iterated Covariances, we have

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = E[\text{Cov}(\hat{\alpha}, \hat{\beta}|\mathbf{X})] + \text{Cov}[E(\hat{\alpha}|\mathbf{X}), E(\hat{\beta}|\mathbf{X})]. \quad (78)$$

We know that $E(\hat{\alpha}|\mathbf{X}) = \alpha$ and $E(\hat{\beta}|\mathbf{X}) = \beta$. We also know that $\text{Cov}(\hat{\alpha}, \hat{\beta}|\mathbf{X}) = \frac{-\sigma^2 \bar{x}}{S_{xx}}$, by Theorem 11.3.3 (Casella and Berger § 11.3). Thus,

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = E\left[\text{Cov}(\hat{\alpha}, \hat{\beta}|\mathbf{X})\right] + \text{Cov}\left[E(\hat{\alpha}|\mathbf{X}), E(\hat{\beta}|\mathbf{X})\right] \quad (79)$$

$$= E\left[\frac{-\sigma^2 \bar{X}}{S_{XX}}\right] + \text{Cov}(\alpha, \beta) \quad (80)$$

$$= -\sigma^2 E\left[\frac{\bar{X}}{S_{XX}}\right] + 0 \quad (81)$$

$$= -\sigma^2 E\left[\frac{\bar{X}}{S_{XX}}\right]. \quad (82)$$

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